

2 Introduction to electronics

With a basic knowledge of robotics and the various aspects of putting a robot together, it's now time to gain a basic understanding of electronic components and how circuits work.

2.1 Breadboards and PCBs

These are most commonly used for building circuits. PCBs stand for printed circuit boards. Rather than using connecting wire and component terminals to form a closed circuit, it is more efficient to use a firm board with a copper coating. The required conductors are drawn by removing copper from the board with a process called etching using a corrosive solution of Ferric Chloride (FeCl_3).

Breadboards

Instead of PCBs, you could also use breadboards. With breadboards, you don't have to touch the soldering iron. All you need to do is plug component terminals in and out of the board to make or break connections. Breadboards have a lot of holes, connected internally into groups. You just have to insert your component's terminal into these holes – it's as simple as that.

PCB (Printed circuit board)

The motherboard of your PC is a standard example of a PCB. The use of breadboards has its own advantages over the use of PCBs. In addition to not having to solder at all, you could proceed onto building circuits once you've designed them. Also, you can reuse your components once you are done and save a lot of money.

On the other hand, once you solder the components and make the necessary connections on a PCB, they have longer shelf lives, and would sustain jerks and shocks. Therefore, ideally, circuits are first built on a breadboard, tested, remade and the final design is soldered on a PCB for practical use. Try following the two golden rules to make fool proof circuits on a breadboard:

Use single strand wires (wires with single conductor wires within the insulator coating) instead of wires with multiple strands (fine multiple conductor wires). Use wires having lengths as close as possible to the distance between the two holes to be connected.

Still, if you have doubt of connection between two holes, insert a safety pin